PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

On

Quick-Kart (A Online shopping App)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By:

1. Rahul Das (2211985038)
2. Souvik Manna (2211985048)

Supervised By:

Dr. Rajat Takkar (9467026112)
Assistant Professor
Chitkara University, Rajpura

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

Table of Contents

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Quick-Kart (Online Shopping App)** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Rajat Takkar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura　　　　　　　　　　　　　　　　　　Rahul Das

Date: 16-05-2026　　　　　　　　　　　　　University Roll No: 2211985038

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Rajat Takkar
Assistant Professor
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the **Department of Computer Science and Engineering** for providing the opportunity to work on this project. The support, encouragement, and academic environment provided by the department played an important role in the successful completion of this work. The knowledge gained during the course of study and the guidance received from the department helped in developing a clear understanding of the concepts required for this project.

I would also like to thank all the **faculty members of the department** for sharing their valuable knowledge and for providing continuous academic support throughout the duration of this project. Their lectures, suggestions, and discussions helped in strengthening the technical understanding necessary for designing and implementing this system.

I am grateful to my **college** for providing the required facilities, resources, and infrastructure that made the development of this project possible. The availability of laboratories, technical resources, and a supportive learning environment allowed me to work on the project effectively and gain practical experience in software development.

I would also like to acknowledge the support and cooperation of my **project teammates**, whose contributions, ideas, and teamwork helped in completing the project successfully. Working together allowed us to exchange knowledge, solve technical challenges, and improve the overall quality and functionality of the system.

Special thanks are also extended to the **technical staff and laboratory assistants** who helped in providing access to the required systems and resources whenever needed. Their assistance during the development and testing stages was greatly appreciated.

Finally, I would like to thank my **family and friends** for their constant encouragement, motivation, and moral support throughout the development of this project. Their understanding and support helped me remain focused and dedicated to successfully completing this work.

I sincerely acknowledge and appreciate everyone who contributed directly or indirectly to the successful completion of this project. Their support and encouragement played a significant role in achieving this milestone.

<u>ABSTRACT</u>

Quick-Kart is a website where people can shop online easily. It helps people find products buy them and keep track of their orders in an simple way. The goal of **Quick-Kart** is to make shopping easier by adding features that help people and sellers communicate and do business together.

When you use **Quick-Kart** you can look at lots of products and add them to your cart to buy later. You do not have to **create an account** to look at products, which makes it easy for people to use the site. You can compare products check prices and read about them before you decide to buy.

Quick-Kart wants to be transparent and help people trust the site. So it gives you all the details about products like what they're how much they cost and what other people think of them. You can also talk to sellers directly if you have questions about products or delivery. This way you do not have to go through someone to get help.

Quick-Kart also has a way to pay for things and manage your orders. It protects your information. Makes sure only you can see it. When you buy something the site will update your order status. Send you messages so you can keep track of what is happening.

Quick-Kart is a place to shop online because it is easy to use, safe and helps you find what you need. It uses technology like **MongoDB, Express.js, React.js** and **Node.js** to make sure it works well and can handle lots of people using it. **Quick-Kart** makes shopping easier and more efficient which is why it is a good choice, for people who want to buy things on the internet.

1. Introduction to Project

1.1 Project Background

In recent years, online shopping platforms have become an essential part of daily life. Customers prefer **digital shopping** because it provides convenience, saves time, and offers access to a wide variety of products from anywhere. Businesses and local sellers are also adopting **e-commerce platforms** to expand their reach and improve customer engagement. As the demand for online shopping continues to grow, the need for efficient and user-friendly shopping systems has increased significantly.

However, users often face difficulties while using traditional shopping applications. Searching for products, managing carts, tracking orders, and completing secure payments can sometimes be slow and confusing. Sellers and administrators also face challenges in managing product inventories, customer data, and order records efficiently. These issues can reduce user satisfaction and make the shopping experience less organized and **time-consuming**.

Quick Kart is developed to solve these problems by providing a modern and centralized e-commerce platform. The system offers features such as easy product browsing, secure user authentication, smooth order management, and efficient product handling. By using modern web technologies, **Quick Kart** aims to create a faster, safer, and more convenient online shopping experience for both customers and administrators.

1.2 Problem Statement

Despite the growth of online shopping platforms, users still face several challenges while shopping online. Customers often find it difficult to quickly search for products, manage carts and complete payments smoothly. Complex interfaces and slow systems can make the shopping experience confusing and time-consuming.

Another major challenge is product and order management for administrators and sellers. Managing inventories, updating product details, and handling customer orders manually can lead to errors and reduced efficiency. Many existing systems also provide limited features and poor organization.

These problems highlight the need for a modern and centralized e-commerce platform like **Quick Kart** that provides easy product management, secure transactions, smooth order tracking, and a user-friendly shopping experience in a single system.

1.3 Objective of the Project

The main objective of the Quick Kart project is to develop a modern e-commerce platform that provides a smooth, secure, and user-friendly online shopping experience for customers and efficient management tools for administrators.

The specific objectives of the project are:
• To allow users to browse, search, and purchase products easily through an online platform.
• To provide secure user authentication and account management features.
• To enable customers to add products to carts and place orders efficiently.
• To integrate secure payment functionality.
• To provide administrators with tools for managing products, categories, and customer orders.
• To maintain organized records of users, products, and transactions within a centralized system.
• To create a responsive and user-friendly interface for both customers and administrators.

By achieving these objectives, Quick Kart aims to simplify the online shopping process and improve convenience, efficiency, and customer satisfaction.

1.4 System Overview

Quick Kart is designed as a web-based e-commerce platform that provides a simple and efficient online shopping experience for customers and administrators. The system is developed using the MERN stack, which includes **MongoDB, Express.js, React.js, and Node.js**. Secure user authentication is implemented using **JSON Web Tokens (JWT)** to ensure safe access to the platform.

After logging into the system, users can browse products, search for items, add products to their cart, and place orders easily through the application. The platform provides a responsive and user-friendly interface that allows customers to view product details and manage their shopping activities smoothly. Secure payment functionality is also integrated to provide safe and reliable transactions.

The system also includes an admin panel for efficient management of products, categories, users, and customer orders. Administrators can add, update, or remove products and maintain organized records within the platform. Additional features such as product categorization, search functionality, and cart management help improve the overall shopping experience and system efficiency.

1.5 Expected Benefits

The **Quick Kart** platform is designed to improve the online shopping experience for both customers and administrators. By providing all shopping and management features within a single system, the platform reduces complexity and makes the shopping process faster and more organized.

The system helps customers easily browse products, **manage carts**, and complete secure purchases with convenience. It also improves efficiency for administrators by simplifying product management, customer handling, and order processing through a centralized platform.

Overall, **Quick Kart** provides a modern and user-friendly approach to e-commerce, making online shopping more convenient, secure, and efficient for users while improving business management and customer satisfaction.

2. Methodology

The **Quick Kart** system is designed using a modular architecture where different components work together to provide a smooth and efficient online shopping experience. The architecture divides the system into multiple modules such as **authentication, product management, cart management, payment handling, and admin management**. Each module performs a specific task, which improves system efficiency, scalability, and maintainability.

The modular design also makes it easier to update or improve individual components without affecting the entire system. This approach ensures that the **Quick Kart** platform operates in a structured and organized manner while providing reliable e-commerce services to users.

2.1 System Approach

The system follows a web-based architecture where the frontend interacts with the backend server through APIs. The backend processes requests, manages user data, and handles product and order operations.

The major modules of the system include:

- User Authentication Module
- Product Management Module
- Cart Management Module
- Payment Module
- Admin Management Module
- User and Order Management Module

2.2 Tools and Technologies

The following technologies were used to develop the Quick-Kart platform.

- **Frontend:** React.js is used to build a dynamic and responsive user interface.
- **Backend:** Node.js and Express.js are used to handle server operations and API development.
- **Authentication:** JSON Web Token (JWT) is used for secure user authentication and authorization.
- **Database:** MongoDB is used to store user data, uploaded documents, flashcards, and quiz results.
- **User Interface:** Tailwind CSS is used to design a modern and responsive interface.
- **Development Environment:** Visual Studio Code is used as the primary development tool.

2.3 Flow of the System

The workflow of the Quick-Kart system is as follows:

1. User registers and logs into the system.
2. The user browses and searches for products.
3. Products are added to the shopping cart.
4. The user places an order and completes payment securely.
5. The system stores product, user, and order details in the database.
6. The administrator manages products, users, and customer orders through the admin.

3. Design

3.1) Use Case Diagram:

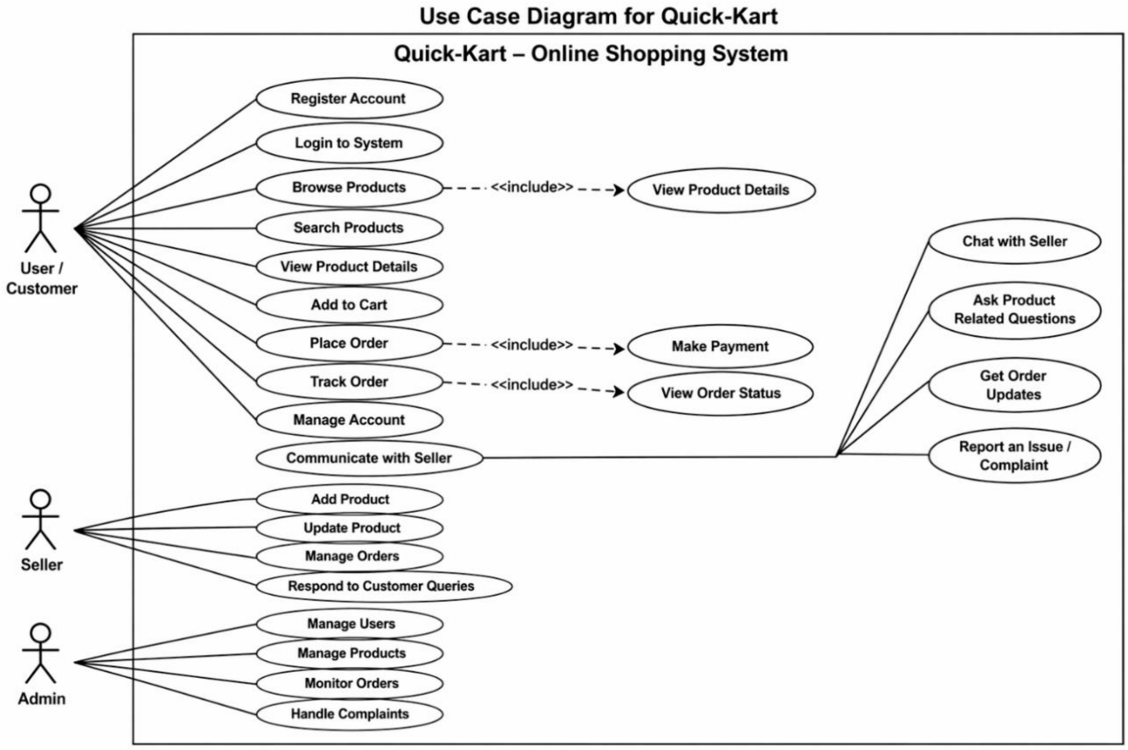


Fig: 3.1

3.2) System Architecture Diagram:

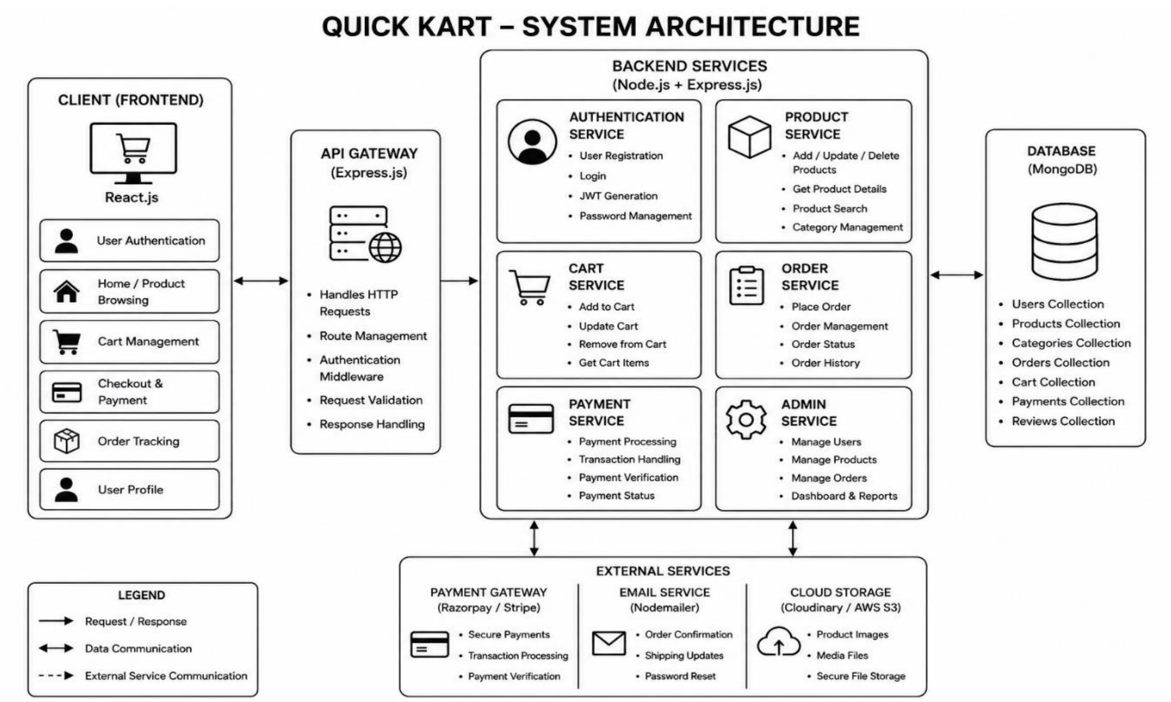


Fig: 3.2

3.3) System Flowchart Diagram:

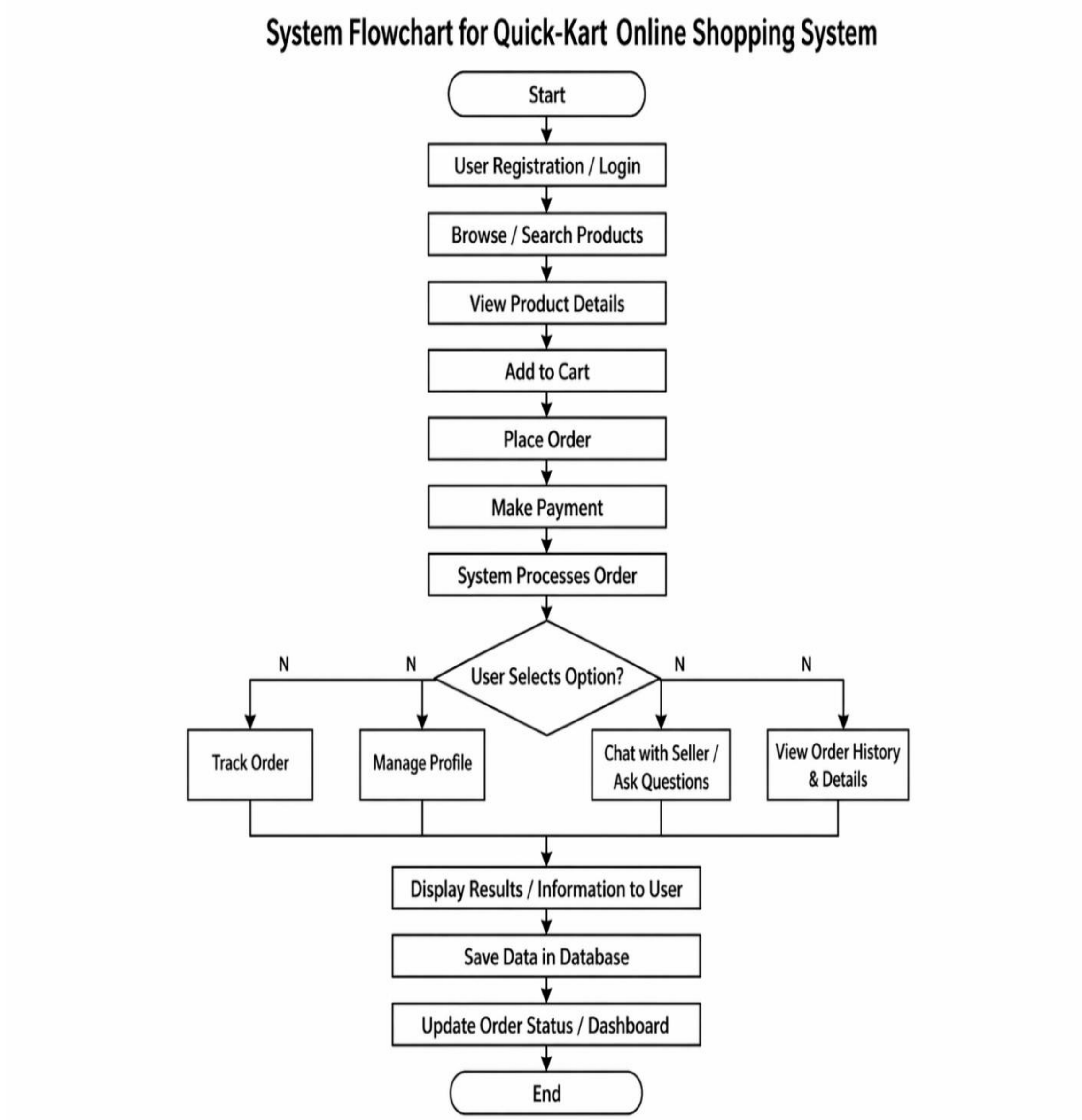


Fig: 3.3

3.4) UML Class Diagram:

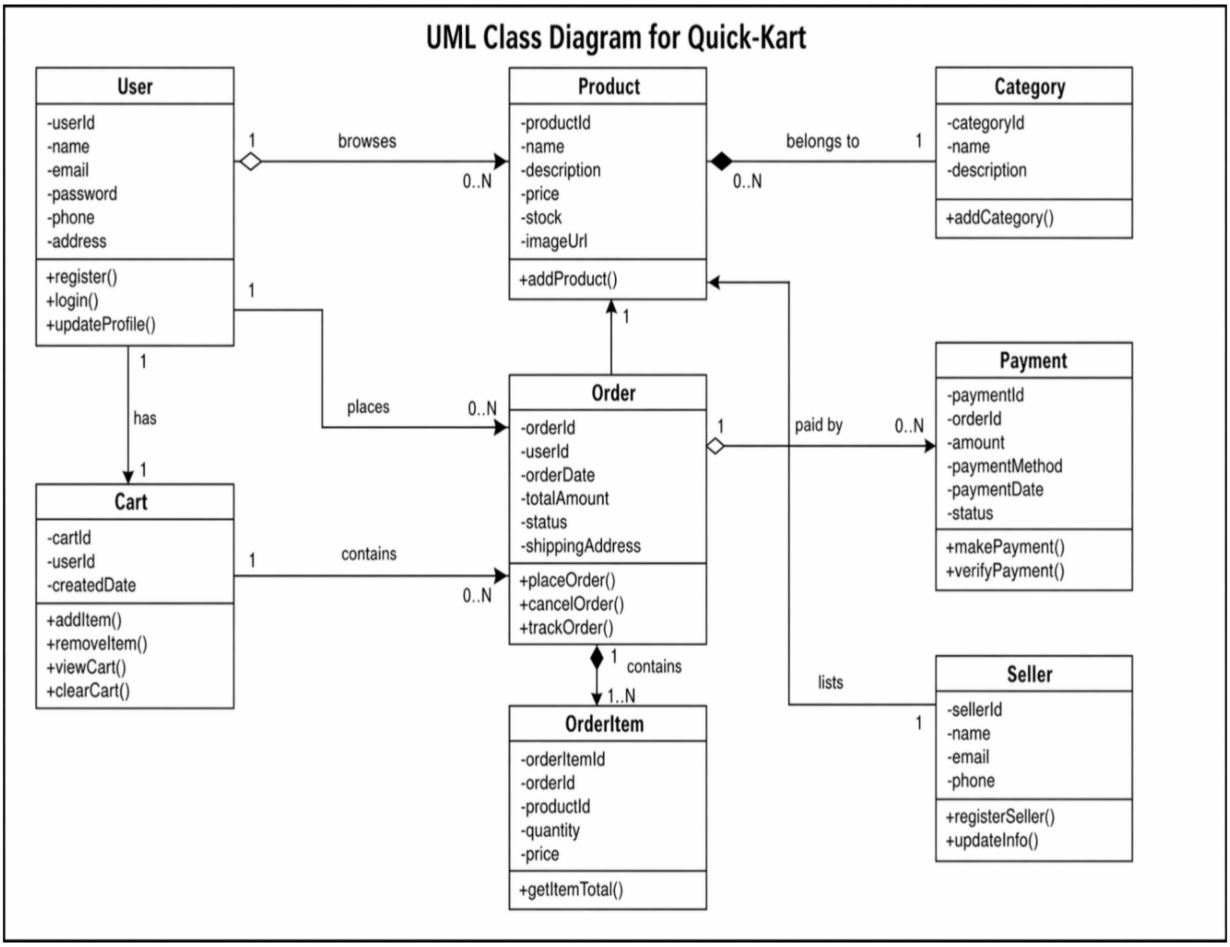


Fig: 3.4

4. Implementation and Results

The Quick-Kart platform was successfully implemented using modern web development technologies.

4.1 Implementation

The system includes several modules:

- **User Authentication:** Secure login and registration are implemented using JWT authentication.
- **Product Management**: Administrators can add, update, and remove products from the system.
- **Product Browsing:** Users can browse products, search items, and view product details easily.
- **Cart Management:** Customers can add products to the cart, update quantities, and remove items.
- **Payment System:** Secure payment functionality is integrated for smooth online transactions.
- **Admin Dashboard:** The dashboard allows administrators to manage users, products, and customer orders efficiently.
- **Database Management:** MongoDB is used to store user details, product information, cart data, and order records.

4.2 Results

The system successfully demonstrates the implementation of a modern e-commerce platform with efficient shopping and management features.

The platform allows users to browse products, manage carts, and complete purchases in a simple and user-friendly way. Screenshots of the working system include the login page, homepage, product page, cart interface, payment page, and admin dashboard.

These results confirm that Quick Kart provides a functional, secure, and efficient online shopping environment for both customers and administrators.

5. Coding

5.1 Db.js

```
const mongoose = require('mongoose');
const connectDB= async () => {
    try {
        const conn = await
mongoose.connect(process.env.MONGO_URI);
        console.log(`MongoDB Connected: ${conn.connection.host}`);
    } catch (error) {
        console.error(`Error: ${error.message}`);
        process.exit(1);
    }
}

module.exports = connectDB;
```

5.2 CartRoutes.js

```
const cartRoute= require('express').Router();
const isAuthenticated = require('../config/auth.js');
const { addToCart, updateCartQuantity,
getCartItems, removeFromCart } =
require('../controllers/cartController');


cartRoute.post('/add-to-cart', isAuthenticated,
addToCart)
cartRoute.put('/update-quantity', isAuthenticated,
updateCartQuantity);
cartRoute.get('/cart-items', isAuthenticated,
getCartItems);
cartRoute.delete('/remove-from-
cart/:productId',isAuthenticated,removeFromCart)

module.exports = cartRoute;
```

5.3 auth.js

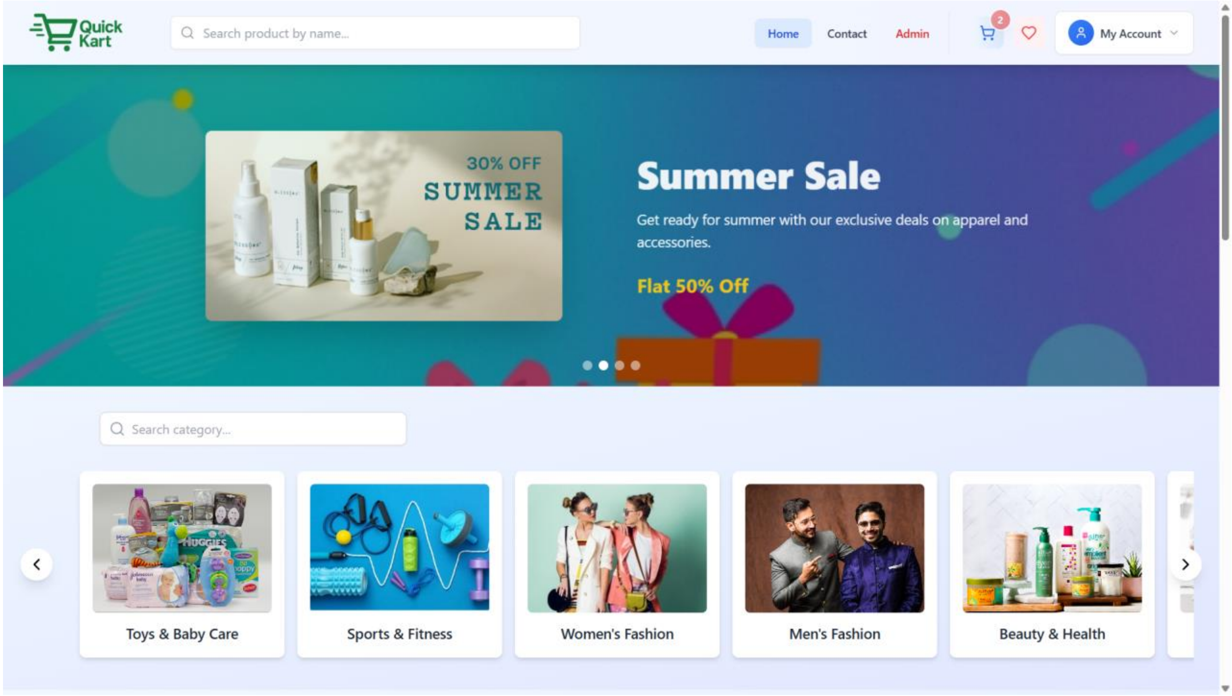
```
const jwt = require('jsonwebtoken');
const dotenv = require('dotenv');
dotenv.config();

const isAuthenticated = (req, res, next)
=> {
    const authHeader =
req.headers.authorization;


    if (!authHeader ||
!authHeader.startsWith('Bearer ')) {
        return res.status(401).json({ success:
false, message: 'Authorization token
missing or malformed' });
    }

    const token = authHeader.split(' ')[1];

    try {
        const decoded = jwt.verify(token,
process.env.JWT_SECRET);
         req.userId = decoded.id;
        next();
    } catch (error) {
        console.error('Token verification
failed:', error);
        return res.status(401).json({ success:
false, message: 'Invalid or expired token'
});
    }
};
```

5.4 userRout.js

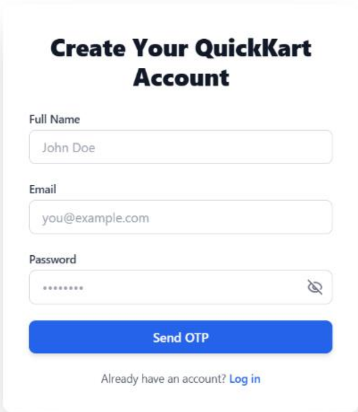
```
constuserRoute=
require('express').Router();
constisAuthenticated =
require('../config/auth.js');
const isAdmin =
require('../config/isAdmin.js')

const {registerUser,loginUser, sendOtp,
getUserDetails, sendLink, resetPassword,
updateAddress,
        uploadProfileImage, getAllUsers,
getUserById, updateUserRole, deleteUser
}= require('../controllers/userController')

userRoute.post('/register', registerUser);
userRoute.post('/login', loginUser);
userRoute.post('/send-otp', sendOtp);
userRoute.get('/me', isAuthenticated,
getUserDetails)
userRoute.post('/forgot-password',
sendLink)
userRoute.post('/reset-
password',resetPassword )
userRoute.put('/update-
address',isAuthenticated, updateAddress)
userRoute.put('/update-
avatar',isAuthenticated,
uploadProfileImage)
userRoute.get('/all-users', isAuthenticated,
isAdmin, getAllUsers)
userRoute.get('/get-user-by-id/:id',
getUserById);
userRoute.put('/:id/role',isAuthenticated,
isAdmin, updateUserRole);
userRoute.delete('/:id/delete',
isAuthenticated, isAdmin, deleteUser);

module.exports = userRoute;
```

7. Prototype Photographs for User Side

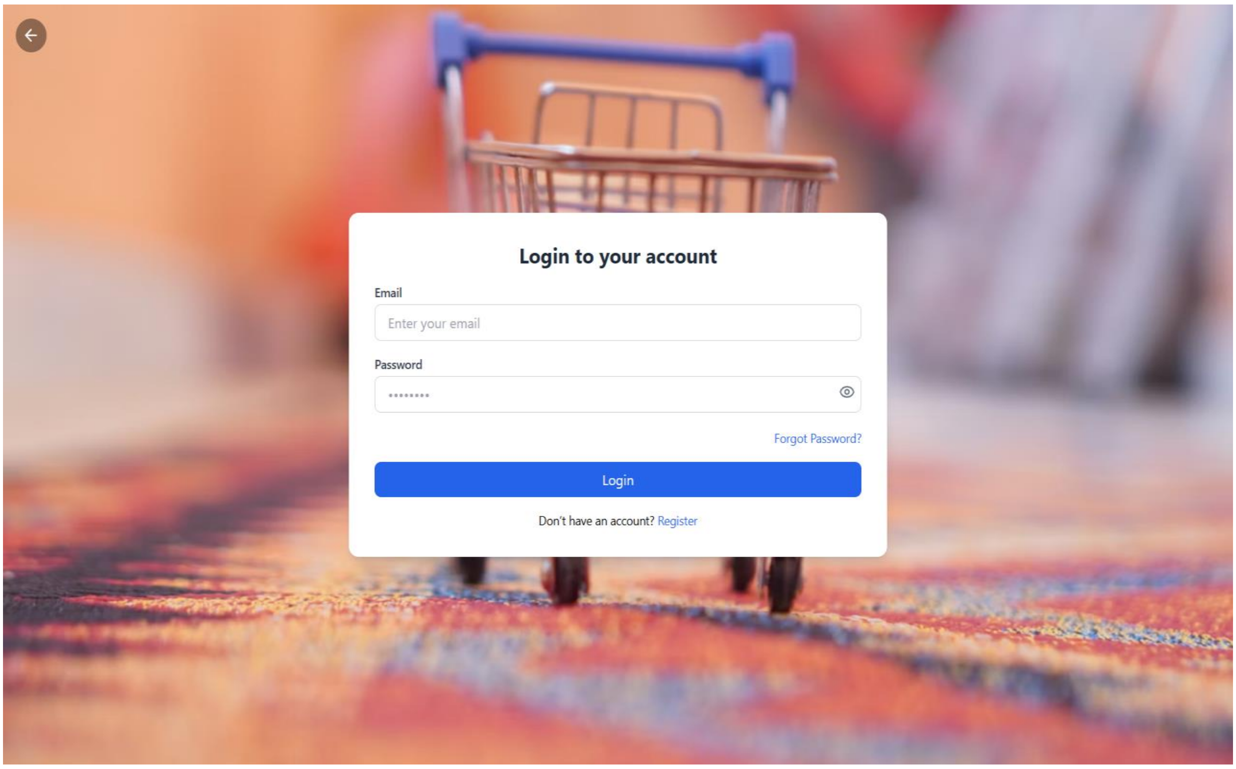
7.1) Quick-Kart Landing Page:



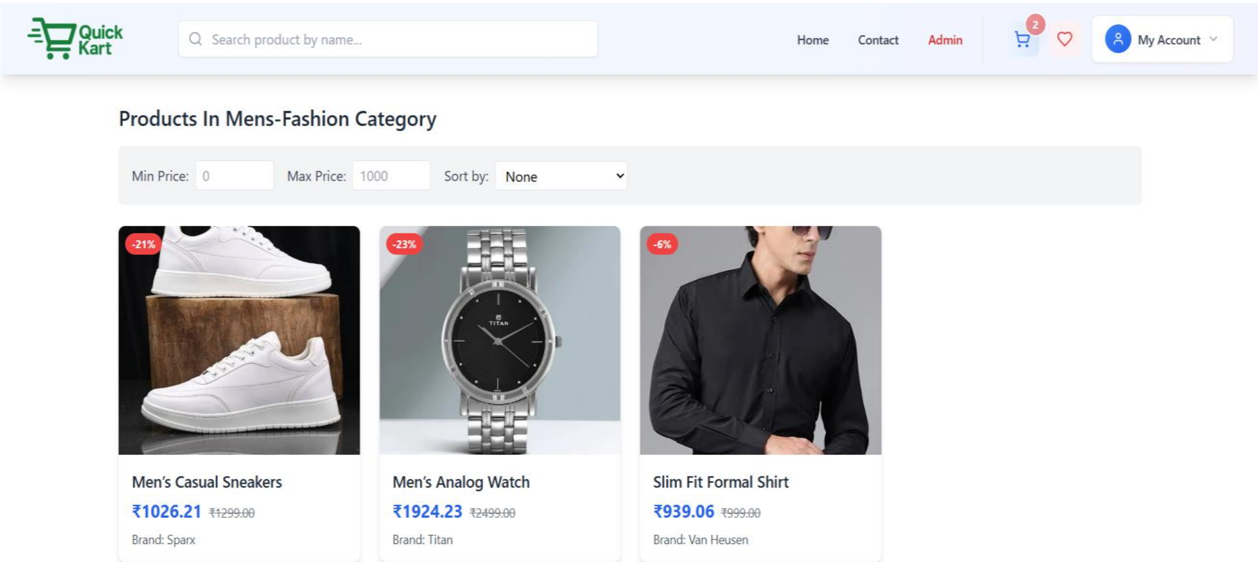
7.2) Quick-Kart User Registration & Login Page：



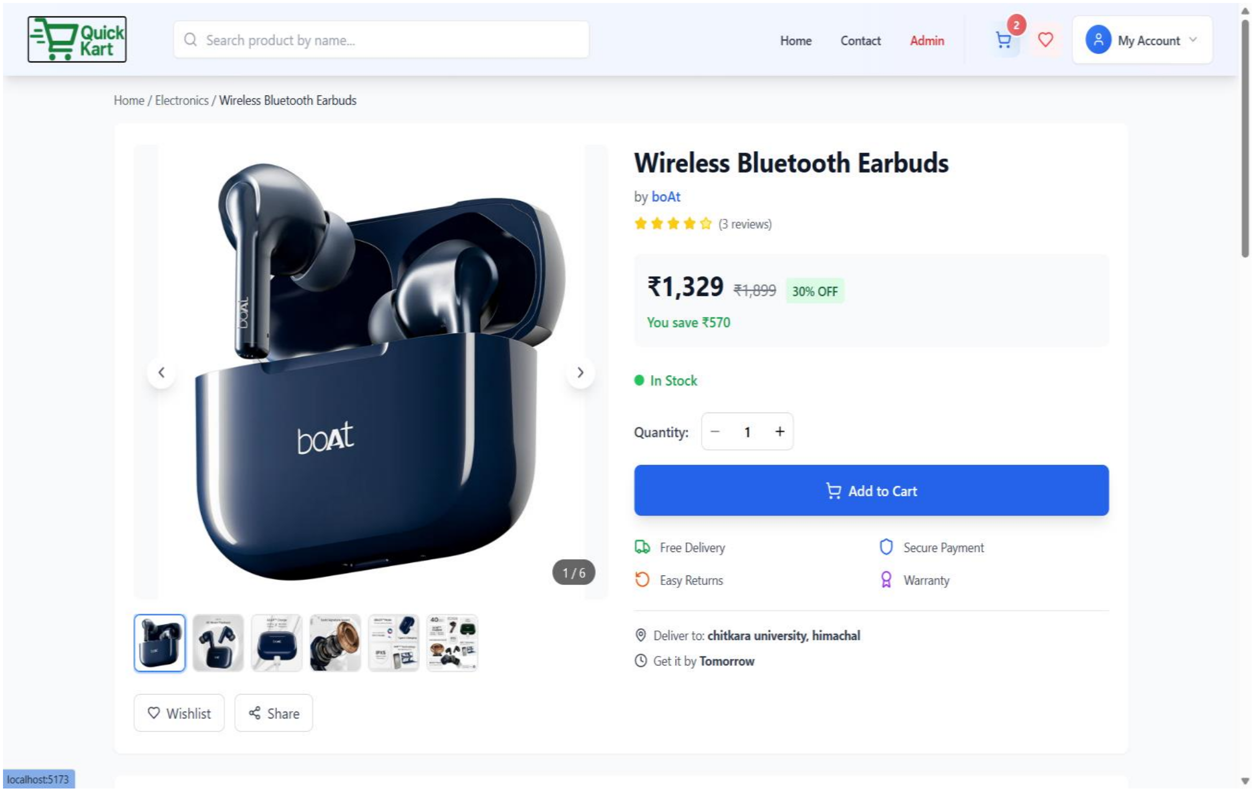
7.3) Login Account after creating Account



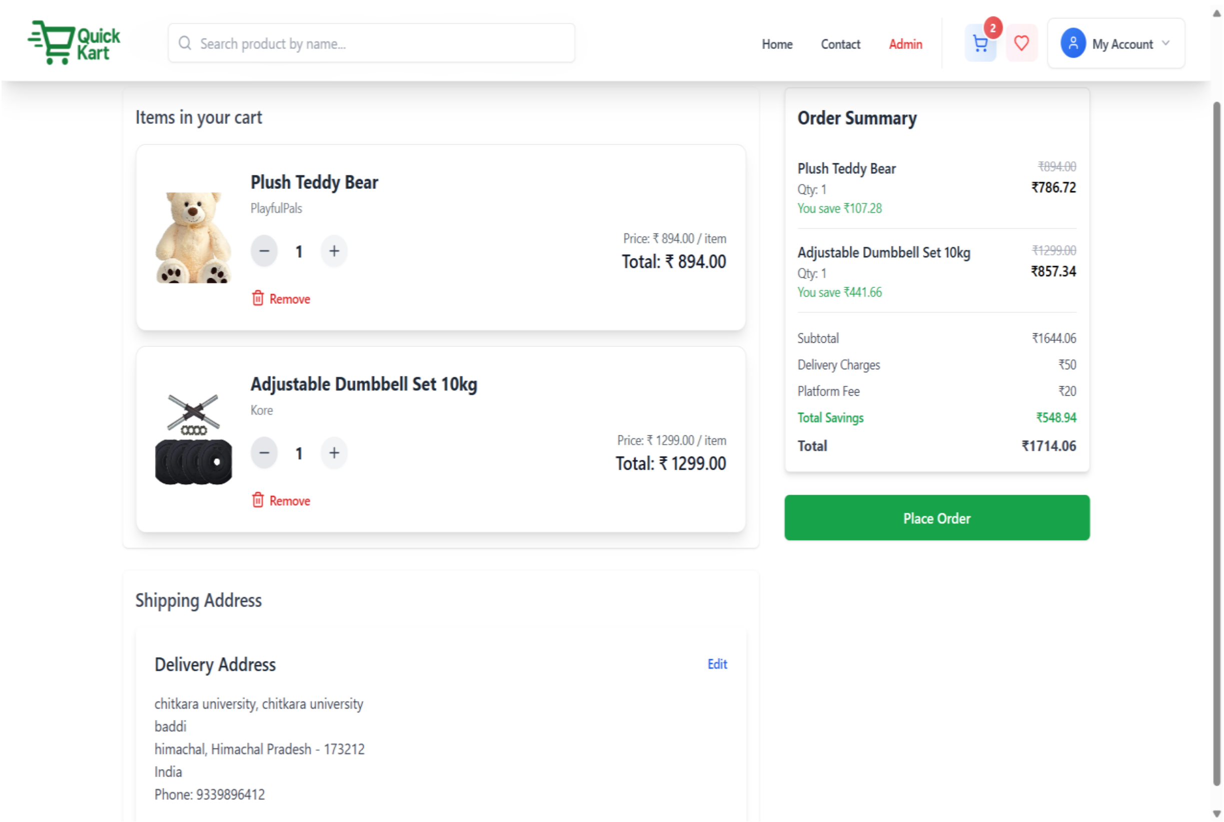
7.4) View Products base on Category :



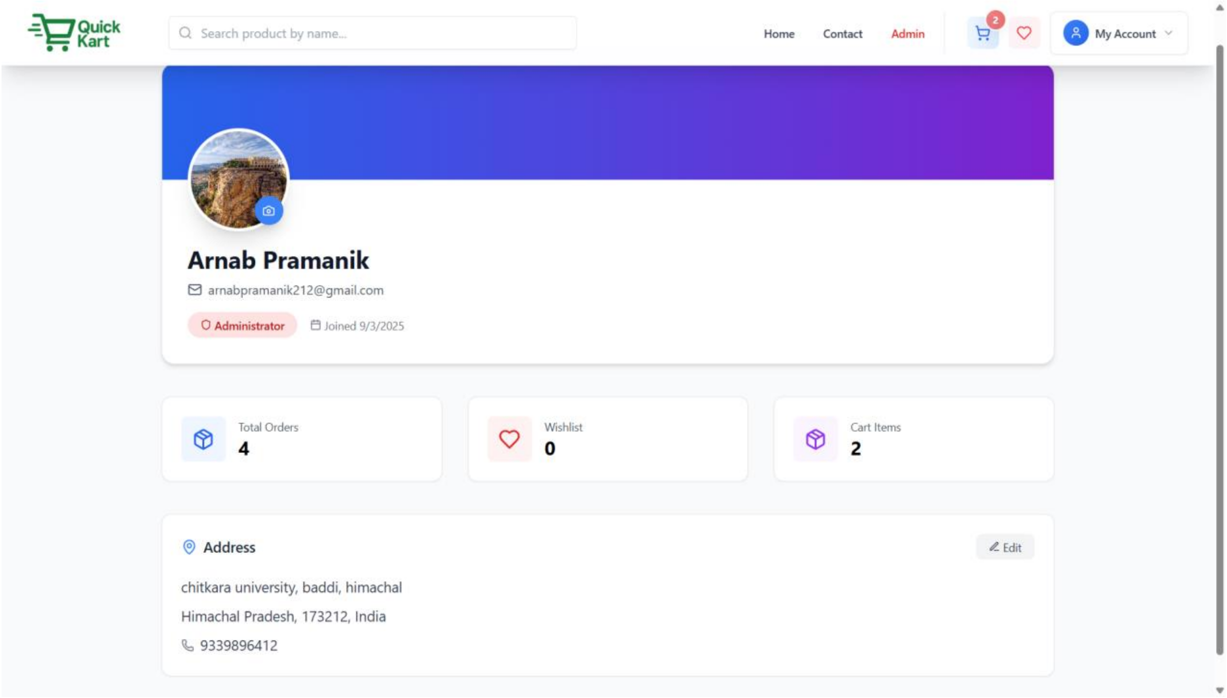
7.5) View Product:

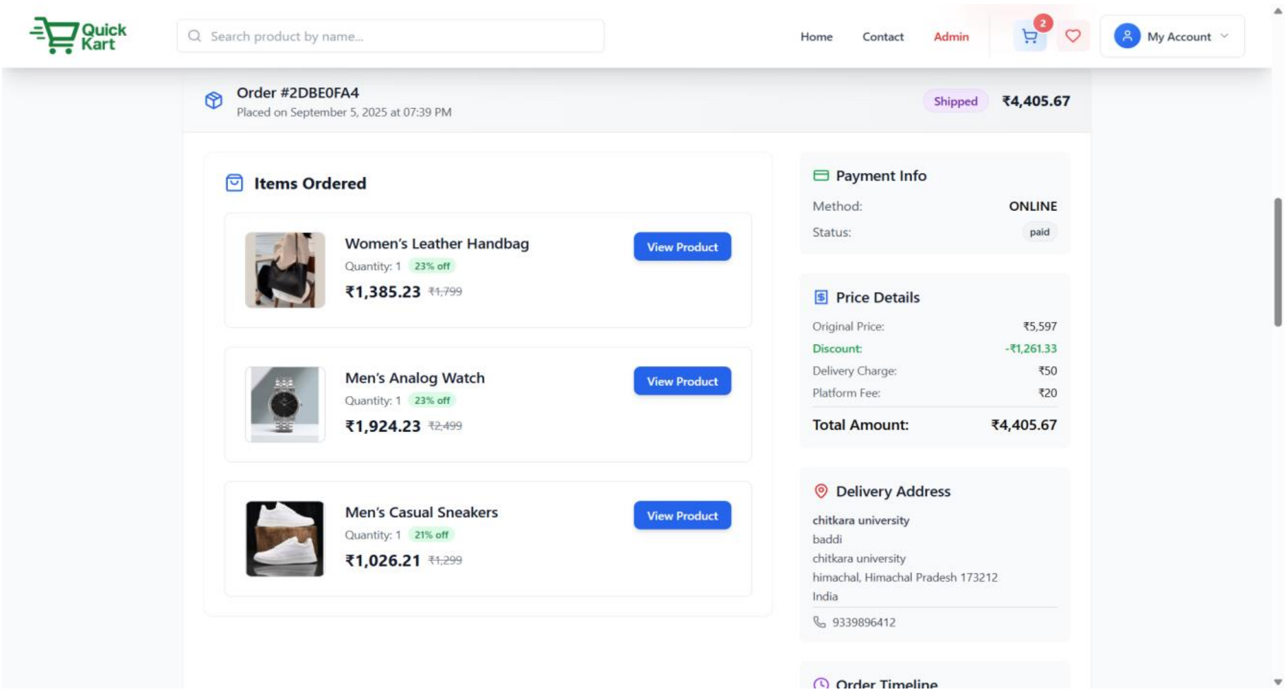


7.6) Cart / Payment page:

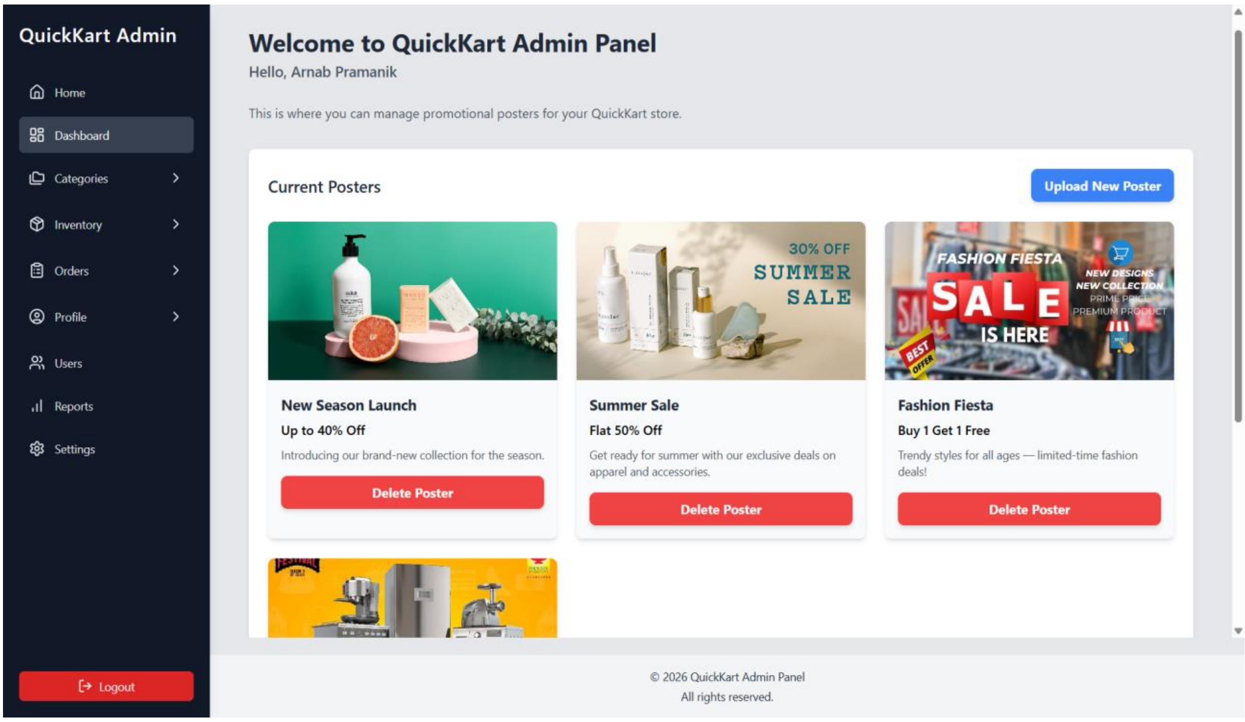


7.7) Profile:

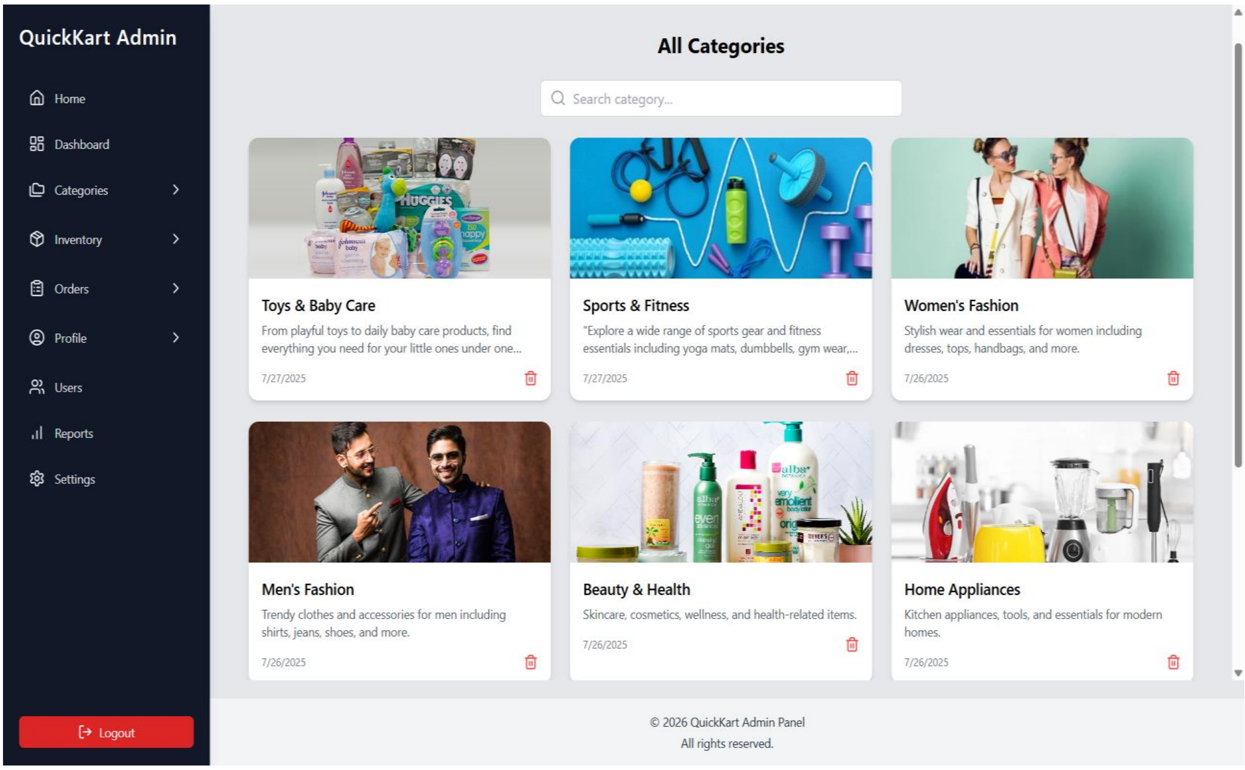

Quick Kart
Search product by name...
Home
Contact
Admin
2
My Account
Arnab Pramanik
arnabpramanik212@gmail.com
Administrator
Joined 9/3/2025
Total Orders
4
Wishlist
0
Cart Items
2
Address
Edit
chitkara university, baddi, himachal
Himachal Pradesh, 173212, India
9339896412


Quick Kart
Search product by name...
Home
Contact
Admin
2
My Account
Order #2DBE0FA4
Placed on September 5, 2025 at 07:39 PM
Shipped
₹4,405.67
Items Ordered
Women's Leather Handbag
Quantity: 1
23% off
₹1,385.23
₹1,799
View Product
Men's Analog Watch
Quantity: 1
23% off
₹1,924.23
₹2,499
View Product
Men's Casual Sneakers
Quantity: 1
21% off
₹1,026.21
₹1,299
View Product
Payment Info
Method:
ONLINE
Status:
paid
Price Details
Original Price:
₹5,597
Discount:
-₹1,261.33
Delivery Charge:
₹50
Platform Fee:
₹20
Total Amount:
₹4,405.67
Delivery Address
chitkara university
baddi
chitkara university
himachal, Himachal Pradesh 173212
India
9339896412
Order Timeline

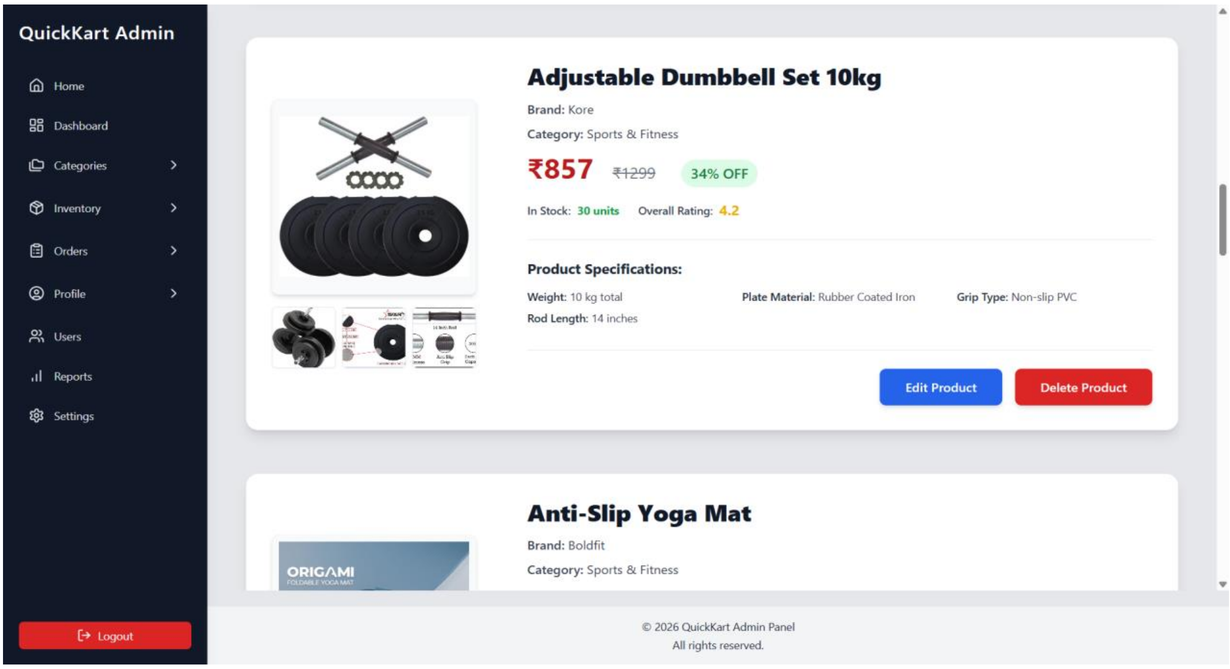
7.8) Admin Dashboard:



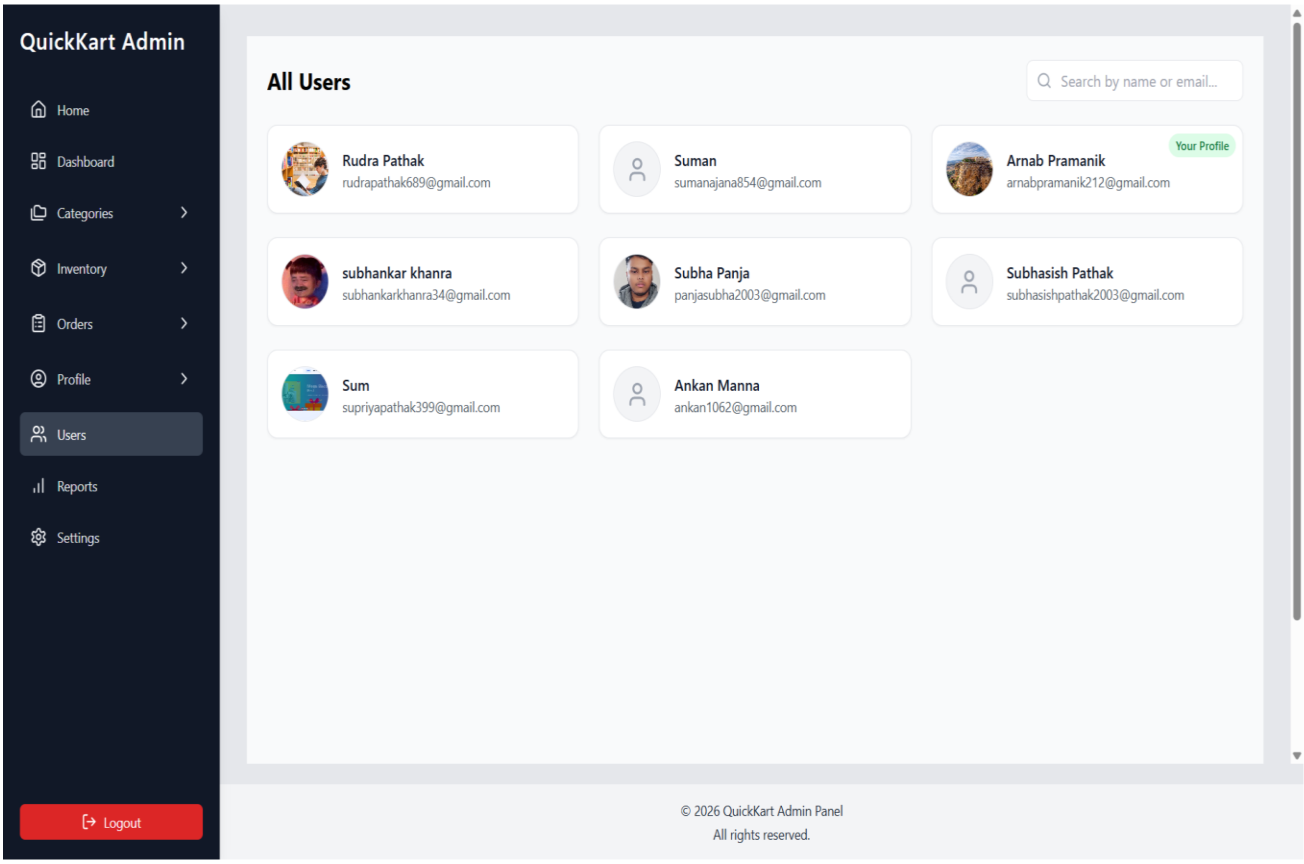
7.8) View Categories in Admin Side:



7.9) View Inventorys:



7.10) View All Users



8. Conclusion and Future Work

8.1 Conclusion

The Quick Kart project successfully demonstrates the development of a modern e-commerce platform using web technologies. By integrating features such as product management, secure authentication, cart management, and online payment functionality, the system provides a smooth and user-friendly shopping experience for customers.

The platform helps simplify the online shopping process and improves efficiency for both users and administrators. Customers can easily browse and purchase products, while administrators can efficiently manage products, users, and orders through a centralized system.

The combination of modern web technologies and responsive design enables Quick Kart to provide a secure, reliable, and efficient online shopping environment.

8.2 Future Work

Several improvements can be implemented in future versions of the system.

Possible future enhancements include:

• Integration of AI-based product recommendation systems for personalized shopping experiences.
• Development of a mobile application for improved accessibility and convenience.
• Addition of customer review and rating features for products.
• Implementation of real-time notifications for offers, payments, and order updates.
• Integration of multiple payment gateways and advanced security features.
• Addition of analytics and reporting tools for administrators to monitor sales and customer activity.

These improvements can further enhance the effectiveness of the **Quick-Kart** platform.

9. Bibliography and References

9.1 Websites

1. https://react.dev/ – Official React documentation for building user interfaces.

2. https://nodejs.org/en/docs – Official Node.js documentation for backend development.

3. https://expressjs.com – Express.js framework documentation for building web APIs.

4. https://www.mongodb.com/docs – MongoDB documentation for database management.

5. https://tailwindcss.com/docs – Tailwind CSS documentation for responsive UI design.

6. https://developer.mozilla.org – MDN Web Docs for JavaScript and web development references.

7. https://vercel.com/docs – Vercel deployment documentation for hosting web applications.

9.2 Books

- Learning React by Kirupa Chinnathambi, Addison-Wesley (2nd Edition) – Provides guidance on building modern web applications using React.

- JavaScript: The Good Parts by Douglas Crockford, O'Reilly Media – Explains important concepts of JavaScript used in frontend and backend development.

- Web Development with Node and Express by Ethan Brown, O'Reilly Media – Describes backend development using Node.js and Express.

9.3 Journals

- “Artificial Intelligence in Education: Applications and Future Prospects,” *IEEE Access*, 2023.

- “AI-Based Intelligent Learning Systems for Digital Education,” *International Journal of Educational Technology in Higher Education*, 2022.

- “Natural Language Processing Techniques for Document Understanding,” *IEEE Transactions on Knowledge and Data Engineering*, 2021.